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United States Patent	12400958
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Hyung; Yong Woo

Method for forming semiconductor device having an air gap between a contact pad and a sidewall of contact hole

Abstract

Disclosed are a semiconductor device and a method for preparing a semiconductor device. The semiconductor device is provided with contact pad structures in contact holes. Each of the contact pad structures is configured to comprise a first contact pad, a second contact pad adaptively covering the first contact pad, and a contact plug located on the second contact pad. The first contact pad is in full contact with an active region in a substrate. In addition, an air gap is formed between the first contact pad and a side wall on a side of the respective contact hole.

Inventors:	Hyung; Yong Woo (Fujian, CN)
Applicant:	Fujian Jinhua Integrated Circuit Co., Ltd. (Fujian, CN)
Family ID:	1000008782878
Assignee:	Fujian Jinhua Integrated Circuit Co., Ltd. (Quanzhou, CN)
Appl. No.:	18/743155
Filed:	June 14, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240339405 A1	Oct. 10, 2024

Foreign Application Priority Data

CN	202011338461.1	Nov. 25, 2020
CN	202022770473.3	Nov. 25, 2020

Related U.S. Application Data

Publication Classification

Int. Cl.: H01L21/768 (20060101); H01L23/528 (20060101); H10D62/10 (20250101)

U.S. Cl.:

CPC H01L23/5283 (20130101); H01L21/76895 (20130101); H10D62/115 (20250101);

Field of Classification Search

CPC: H01L (21/28525); H01L (21/76895); H01L (21/76897); H01L (23/5283); H10D (62/115); H10B (12/0335); H10B (12/315); H10B (12/482)

USPC: 438/262

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Primary Examiner: Pham; Hoai V

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a divisional application of U.S. patent application Ser. No. 17/521,849, which claims the priority of Chinese patent application No. 202022770473.3, filed on Nov. 25, 2020 and entitled “Semiconductor device”, and the priority of Chinese patent application No. 202011338461.1, filed on Nov. 25, 2020 and entitled “Semiconductor device and method for preparing semiconductor device”, which are incorporated herein by reference in their entirety.

FIELD OF THE INVENTION

(1) The present disclosure relates to the field of semiconductors, and in particular to a semiconductor device and a method for preparing a semiconductor device.

BACKGROUND OF THE INVENTION

(2) With the development of technology, the level of integration of a semiconductor device gradually increases in order to satisfy a requirement for miniaturization of the semiconductor device. The increase of the level of integration requires a reduction in the dimension of a contact hole in the semiconductor device. A smaller contact hole will have a larger aspect ratio, and it will be difficult to control a morphology of a bottom part of the contact hole during a preparation

process, causing deformation and oxidization of the contact hole. As a result, when a contact pad structure is formed in a contact hole having a larger aspect ratio, a bottom part of the contact pad structure and an active region exposed by the contact hole have poor contact, and the conductivity is relatively poor, thereby greatly affecting performance of a semiconductor device.

SUMMARY OF THE INVENTION

(3) The technical problem to be solved by the present invention is how to realize effective contact between a contact pad structure and an active region in a semiconductor device with a contact hole having a larger aspect ratio to greatly improve a conductivity characteristic of the semiconductor device.

(4) In order to solve the above-mentioned technical problem, the present disclosure provides a semiconductor device and a method for preparing a semiconductor device.

(5) The first aspect of the present disclosure provides a semiconductor device, which may include: a substrate including at least an active region and a shallow trench isolation region; bit line structures located on the substrate and extending in a first direction; isolation fences located between adjacent bit line structures and arranged at intervals in the first direction; contact holes located in an area defined by adjacent bit line structures in a second direction and adjacent isolation fences in the first direction, where the second direction is perpendicular to the first direction, a bottom part of each of the contact holes extends into the substrate and at least exposes part of the active region and part of the shallow trench isolation region; and a contact pad structure located in a respective contact hole, and including a first contact pad, a second contact pad adaptively covering the first contact pad, and a contact plug located on the second contact pad, where an air gap is formed between the first contact pad and a side wall on a side of the respective contact hole.

(6) The second aspect of the present disclosure provides a method for preparing a semiconductor device, which may include: forming, on a substrate, bit line structures extending in a first direction, where the substrate includes at least an active region and a shallow trench isolation region, the bit line structures are arranged at intervals in a second direction, and the second direction is perpendicular to the first direction; etching the substrate between adjacent bit line structures to form a contact opening that extends into the substrate and at least expose part of the active region and part of the shallow trench isolation region; forming isolation fences in the contact opening, wherein the isolation fences are arranged at intervals in the first direction to isolate the contact opening into a plurality of contact holes; performing, by using a germane gas, pretreatment on a bottom part of each of the contact holes; forming a first contact pad in a respective contact hole that has been subjected to the pretreatment; forming, on the first contact pad, a second contact pad adaptively covering the first contact pad, where an air gap is formed between the first contact pad and a side wall on a side of the respective contact hole; and forming, in the contact holes, contact plugs, respectively.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The scope of the present disclosure will be better understood from reading the following detailed description of exemplary embodiments in conjunction with the accompanying drawings. In the accompanying drawings:

(2) FIG. 1 is a schematic top structural view of a semiconductor device provided in an embodiment of the present disclosure;

(3) FIG. 2A is a schematic cross-sectional structural view along a line A-A' in FIG. 1 as provided in an embodiment of the present disclosure;

(4) FIG. 2B is a schematic cross-sectional structural view along the line A-A' in FIG. 1 as provided in another embodiment of the present disclosure;

- (5) FIG. 2C is a schematic cross-sectional structural view along the line A-A' in FIG. 1 as provided in another embodiment of the present disclosure;
- (6) FIG. 3 is a schematic cross-sectional structural view along a line B-B' in FIG. 1 as provided in an embodiment of the present disclosure;
- (7) FIG. 4 is a schematic flowchart of a method for preparing a semiconductor device as provided in an embodiment of the present disclosure;
- (8) FIG. 5A is a schematic top structural view of a substrate **10**;
- (9) FIG. 5B is a schematic cross-sectional structural view along a line A-A' in FIG. 5A;
- (10) FIG. 5C is a schematic cross-sectional structural view along the line B-B' in FIG. 5A;
- (11) FIG. 6A is a schematic top structural view of a semiconductor device provided with bit line structures;
- (12) FIG. 6B is a schematic cross-sectional structural view along a line A-A' in FIG. 6A;
- (13) FIG. 6C is a schematic cross-sectional structural view along the line B-B' in FIG. 6B;
- (14) FIG. 7A is a schematic top structural view of a semiconductor device provided with a contact opening;
- (15) FIG. 7B is a schematic cross-sectional structural view along a line A-A' in FIG. 7A;
- (16) FIG. 7C is a schematic cross-sectional structural view along a line B-B' in FIG. 7A;
- (17) FIG. 8A is a schematic top structural view of a semiconductor device provided with contact holes **15**;
- (18) FIG. 8B is a schematic cross-sectional structural view along a line A-A' in FIG. 8A;
- (19) FIG. 8C is a schematic cross-sectional structural view along a line B-B' in FIG. 8A;
- (20) FIG. 9A is a schematic cross-sectional structural view, along the line A-A', of a semiconductor device provided with first contact pads **161**;
- (21) FIG. 9B is a schematic cross-sectional structural view, along the line B-B', of a semiconductor device provided with first contact pads **161**;
- (22) FIG. 10A is a schematic cross-sectional structural view, along the line A-A', of a semiconductor device provided with second contact pads **162**;
- (23) FIG. 10B is a schematic cross-sectional structural view, along the line B-B', of a semiconductor device provided with second contact pads **162**;
- (24) FIG. 11A is a schematic top structural view of a semiconductor device provided with contact plugs **163**;
- (25) FIG. 11B is a schematic cross-sectional structural view along a line A-A' in FIG. 11A; and
- (26) FIG. 11C is a schematic cross-sectional structural view along a line B-B' in FIG. 11A.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(27) In order to make the objects, technical solutions and advantages of the present disclosure clearer, the implementations of the present disclosure will be described in details below in conjunction with the accompanying drawings and embodiments, so as to fully understand and implement the implementation process of how to apply technical means to solve technical problems and achieve technical effects.

(28) With the development of technology, the level of integration of a semiconductor device gradually increases in order to satisfy a requirement for miniaturization of the semiconductor device. The increase of the level of integration requires a reduction in the dimension of a contact hole in the semiconductor device. A smaller contact hole will have a larger aspect ratio, and it will be difficult to control a morphology of a bottom part of the contact hole during a preparation process, causing deformation and oxidization of the contact hole. As a result, when a contact pad structure is formed in a contact hole having a larger aspect ratio, a bottom part of the contact pad structure and an active region exposed by the contact hole have poor contact, and the conductivity is relatively poor, thereby greatly affecting performance of a semiconductor device.

(29) Thus, the present disclosure provides a semiconductor device and a method for preparing a semiconductor device. The semiconductor device is provided with contact pad structures in contact

holes. Each of the contact pad structures is configured to include a first contact pad, a second contact pad adaptively covering the first contact pad, and a contact plug located on the second contact pad. The first contact pad is in full contact with an active region in a substrate, which can effectively reduce contact resistance. Moreover, by providing contact pads in the form of a stacking structure, each of the first contact pads can function as a buffer layer for a respective second contact pad so as to avoid doped ions in the second contact pad from penetrating into the substrate and affecting conductivity. In addition, an air gap in each contact hole is formed between the first contact pad and a side wall on a side of the contact hole, which can effectively reduce an influence of the air gap on the conductivity, thereby facilitating significant improvement of a conductivity characteristic of the semiconductor device.

Embodiment One

(30) With reference to FIG. 1, there is shown a schematic top structural view of a semiconductor device provided in an embodiment of the present disclosure. The semiconductor device includes a substrate **10**, bit line structures **13**, isolation fences **14**, contact holes and contact pad structures **16**.

(31) The substrate **10** includes at least an active region **11** and a shallow trench isolation region **12**.

(32) The bit line structures **13** are located on the substrate **10** and extend in a first direction.

(33) The isolation fences **14** are located between adjacent bit line structures **13** and arranged at intervals in the first direction.

(34) The contact holes **15** are located in an area defined by adjacent bit line structures **13** in a second direction and adjacent isolation fences **14** in the first direction. The second direction is perpendicular to the first direction. A bottom part of each of the contact holes **15** extends into the substrate **10** and at least exposes part of the active region **11** and part of the shallow trench isolation region **12**.

(35) The contact pad structures **16** are located in the respective contact holes **15**. Each of the contact pad structures **16** includes a first contact pad **161**, a second contact pad **162** adaptively covering the first contact pad **161**, and a contact plug **163** located on the second contact pad **162**. An air gap **17** is formed between the first contact pad **161** and a side wall on a side of the respective contact hole **15**.

(36) The substrate **10** may include a semiconductor substrate. As a specific example, the substrate **10** may include a silicon substrate, a silicon-germanium substrate or a silicon substrate on an insulator. The substrate **10** may include the active region **11** and the shallow trench isolation region **12**, and the shallow trench isolation region **12** may define the active region **11** into a plurality of areas.

(37) In the embodiment of the present disclosure, the first direction may be represented by D1, and the second direction perpendicular to the first direction may be represented by D2. The bit line structures **13** extend in the direction D1, and the plurality of bit line structures are arranged at intervals in the direction D2.

(38) As an example, each of the bit line structures **13** may include a bit line **131** and a hard mask layer **132** located on the bit line **131**. Some of the bit line structures **13** may be in contact with the substrate **10** by means of bit line contact plugs **133**. The bit line contact plugs **133** are arranged in respective bit line contact holes **133'** inside the substrate **10**, and each of the bit line contact holes **133'** may have a transverse size greater than that of each of the bit line contact plugs **133**. In addition, bit line spacers **134** may also be provided, each of which may adaptively cover a respective bit line structure **13** and fill in an air gap between a respective bit line contact plug **133** and a respective bit line contact hole **133'**. An upper surface of the bit line contact plug **133** has the same height as an upper surface of the substrate **10**. Referring particularly to FIG. 2A or 2B, FIG. 2A shows a schematic cross-sectional structural view along a line A-A' in FIG. 1 as provided in an embodiment of the present disclosure, and FIG. 2B shows a schematic cross-sectional structural view along the line A-A' in FIG. 1 as provided in another embodiment of the present disclosure. It should be noted that the size of the air gap **17** is enlarged in FIGS. 2A and 2B for convenience of

view, and the size of the air gap **17** is not specifically limited.

(39) As an example, the isolation fences **14** may be formed by depositing an insulation material. Specifically, the isolation fences **14** may be formed by depositing silicon nitride.

(40) It should be noted that the isolation fences **14** may be formed before the contact holes **15** are formed, and may also be formed after the contact holes are formed. In the embodiment of the present disclosure, description is provided as the isolation fences **14** are formed after the contact holes **15** are formed. Referring particularly to FIG. **3**, there is shown a schematic cross-sectional structural view along a line B-B' in FIG. **1** as provided in an embodiment of the present disclosure.

(41) Referring back to FIG. **2A** or **2B**, the contact pad structures **16** are located in the respective contact holes **15**. As an example, pretreatment may be performed on each of the contact holes **15** by using a germane gas before a respective first contact pad **161** is deposited, to remove impurities on an upper surface of the active region **11**, so that the first contact pad **161** and the active region **11** are in good contact, thereby improving conductivity. The first contact pad **161** is then formed, by means of chemical vapor deposition, physical vapor deposition or the like, in a respective contact hole **15** that has been subjected to the pretreatment. It is possible to control, by means of selecting a suitable deposition condition, the first contact pad **161** to be in contact with part of a substrate surface of the respective contact hole **15**. As one example, the first contact pad **161** may be in contact with part of a surface of each of the active region **11** and the shallow trench isolation region **12** in the respective contact hole **15** (not shown in the figures). As another example, the first contact pad **161** may be formed, by a selective epitaxial growth process, on the active region **11** in the respective contact hole **15** that has been subjected to the pretreatment. The first contact pad **161** is only in contact with the active region **11**.

(42) In addition, in the embodiment of the present disclosure, a polygonal first contact pad **161** having a planer top surface may be formed by controlling growth rates in different directions. In another embodiment, it is also possible to form a first contact pad **161** having an air gap between the same and a side wall on a side of a respective contact hole **15** and having an approximately elliptic cross-section. The first contact pad **161** may include a silicon germanium layer, and may also be an n-doped silicon germanium layer.

(43) The second contact pad **162** adaptively covers the first contact pad **161**. As an example, the second contact pad **162** may be formed on an upper surface of the first contact pad **161** by a selective epitaxial growth process, and an upper surface of the second contact pad **162** may be higher than a lower surface of the bit line structure **13**.

(44) The second contact pad **162** and the first contact pad **161** have different chemical compositions. For example, the second contact pad **162** may be formed from a material different from that of the first contact pad **161**, and the second contact pad **162** may also be formed from the same material as that of the first contact pad **161** but with a different ion doping concentration. As a specific example, the second contact pad **162** may include a silicon phosphide layer.

(45) The first contact pad **161** is formed by using the selective epitaxial growth process, and a bottom part of the first contact pad **161** is only in contact with the active region **11**. After the second contact pad **162** is formed by using the selective epitaxial growth process, an air gap is formed between each of the first contact pad **161** and the second contact pad **162** and a side wall on a side of a respective contact hole **15**. In such a structure, the first contact pad **161** may function as a buffer layer for the second contact pad **162**, avoiding doped ions in the second contact pad **162** from penetrating into the substrate and affecting conductivity. Moreover, the air gap **17** in the contact hole **15** is located between the first contact pad **162** and a side wall on a side of the contact hole **15**, which can effectively reduce an influence of the air gap **17** on the conductivity, thereby greatly improving performance of a semiconductor device.

(46) In the embodiment of the present disclosure, since the first contact pad **161** is in contact with part of the substrate surface of the contact hole **15**, an air gap will be formed between the first contact pad **161** and each of a side wall on a side of the respective contact hole **15** and a bottom

surface of the respective contact hole **15** after the first contact pad **161** and the second contact pad **162** are both formed, and such an air gap may not be able to be completely filled when the contact plug **163** is subsequently formed by using a depositing process. As an example, the contact plug **163** may be formed on the second contact pad **162** by means of a chemical vapor deposition process or a physical vapor deposition process. Process conditions are controlled in such a way that the contact plug **163** only covers part of the surface of the respective second contact pad **162**, and the air gap **17** is formed between each of the first contact pad **161** and the second contact pad **162** and the respective contact hole **15**, as depicted in FIG. 2A for details.

(47) As another example, the contact plug **163** may be formed on the respective second contact pad **162** by choosing a depositing process with a better step coverage characteristic, such that the contact plug **163** covers the second contact pad **162** and is in contact with part of the respective first contact pad **161**, the air gap between each of the first contact pad **161** and the second contact pad **162** and the respective contact hole **15** is partially filled, and the air gap **17** is still reserved between the first contact pad **161** and a bottom part of the contact hole **15**, as depicted in FIG. 2B for details.

(48) As an example, the contact plug **163** may comprise a first contact insertion plug **1631** and a second contact insertion plug **1632** located on the first contact insertion plug **1631**. An upper surface of the first contact insertion plug **1631** may be configured to be higher than an upper surface of a bit line **131**, so that a contact interface between the first contact insertion plug **1631** and the second contact insertion plug **1632** is higher than the bit line **131**, thereby avoiding ions at the contact interface from penetrating into the bit line **131** and affecting device performance.

(49) In the embodiment of the present disclosure, the first contact insertion plug **1631** may include a polysilicon layer, a metal silicide or a metal layer, and the second contact insertion plug **1632** may include a metal layer. As a specific example, the first contact insertion plug **1631** may be a silicon phosphide layer, and the second contact insertion plug **1632** may be made of tungsten.

(50) As another example, with reference to FIG. 2C, there is shown a schematic cross-sectional structural view along the line A-A' in FIG. 1 as provided in another embodiment of the present disclosure. The contact pad structure **16** may also include a heavily-doped semiconductor layer **18** located between the first contact insertion plug **1631** and the second contact insertion plug **1632**, and contact resistance between the first contact insertion plug **1631** and the second contact insertion plug **1632** can be effectively reduced by providing the heavily-doped semiconductor layer **18**.

(51) The above embodiment provides a semiconductor device with contact pad structures **16** in respective contact holes **15**. Each of the contact pad structures **16** is configured to include a first contact pad **161**, a second contact pad **162** adaptively covering the first contact pad **161**, and a contact plug **163** located on the second contact pad **162**. The first contact pad **161** is in full contact with an active region **11** in a substrate **10**, which can effectively reduce contact resistance.

Moreover, by providing such contact pads in a stacking structure, the first contact pad **161** can function as a buffer layer for the second contact pad **162** so as to avoid doped ions in the second contact pad **162** from penetrating into the substrate **10** and affecting conductivity. In addition, an air gap **17** in each contact hole **15** is formed between the first contact pad **161**, the second contact pad **162**, the bit line spacer **134**, the shallow trench isolation region **12** and a side wall on a side of the contact hole **15**, which can effectively reduce an influence of the air gap **17** on the conductivity, thereby facilitating significant improvement of a conductivity characteristic of the semiconductor device.

(52) Another aspect of the present disclosure further provides a method for preparing a semiconductor device. For details, reference can be made to the description in the following Embodiment Two.

Embodiment Two

(53) With reference to FIG. 4, there is shown a schematic flowchart of the method for preparing a semiconductor device as provided in an embodiment of the present disclosure, and the method includes the following steps **S401** to **S407**.

(54) At step **S401**, bit line structures **13** extending in a first direction are formed on a substrate **10**, where the substrate **10** includes at least an active region **11** and a shallow trench isolation region **12**, the bit line structures **13** are arranged at intervals in a second direction, and the second direction is perpendicular to the first direction.

(55) At step **S402**, the substrate **10** between adjacent bit line structures **13** is etched to form a contact opening that extends into the substrate **10** and at least expose part of the active region **11** and part of the shallow trench isolation region **12**.

(56) At step **S403**, isolation fences **14** are formed in the contact opening, where the isolation fences **14** are arranged at intervals in the first direction to isolate the contact opening into a plurality of contact holes **15**.

(57) At step **S404**, pretreatment is performed on a bottom part of each of the contact holes **15** by using a germane gas.

(58) At step **S405**, a first contact pad **161** is formed in a respective contact hole **15** that has been subjected to the pretreatment.

(59) At step **S406**, a second contact pad **162** adaptively covering the first contact pad **161** is formed on the first contact pad **161**, where an air gap **17** is formed between the first contact pad **161** and a side wall on a side of the respective contact hole **15**.

(60) At step **S407**, contact plugs **163** are formed in the contact holes **15**, respectively.

(61) With reference to FIGS. 5A and 5B, FIG. 5A shows a schematic top structural view of a substrate **10**, and FIG. 5B shows a schematic cross-sectional structural view along a line A-A' in FIG. 5A. The substrate **10** may include a semiconductor substrate. As an example, the substrate **10** may include a silicon substrate, a silicon-germanium substrate or a silicon substrate on an insulator. The substrate **10** may internally include the active region **11** and the shallow trench isolation region **12**, and the shallow trench isolation region **12** may define the active region **11** into a plurality of areas.

(62) In the embodiment of the present disclosure, the first direction may be represented by **D1**, and the second direction perpendicular to the first direction may be represented by **D2**. In addition, with reference to FIG. 5C, there is shown is a schematic cross-sectional structural view along a line B-B' in FIG. 5A. The substrate **10** may also be internally provided with buried word line structures **19**. The buried word line structures **19** extend in the direction **D2**. Each of the buried word line structures **19** may include a word line **191** and an insulation isolation layer **192** located on the word line **191**, and the buried word line structures **19** may be realized by means of a conventional approach in the art, which will not be described herein for brevity. In addition, dielectric layers **193** may also be arranged outside the buried word line structures **19**, respectively, and the dielectric layers **193** may be oxide layers or nitride layers.

(63) Step **S401** may involve the following processes. Specifically, a hard mask layer is deposited and patterned on the substrate **10**. The patterned hard mask layer is used as a mask to etch the substrate **10**, so as to form bit line contact holes **133'** in the substrate **10**. A conductive material covering the bit line contact holes **133'** is deposited, where an upper surface of the conductive material may be the same height as an upper surface of the substrate **10**. A conductive layer covering the substrate **10** and the conductive material as well as a hard mask layer covering the conductive layer are deposited subsequently. And then the hard mask layer is patterned and the patterned hard mask layer is used as a mask to etch the conductive layer and the conductive material in the bit line contact holes **133'**, so as to form bit line structures **13** extending in the direction **D1**. The patterned conductive layer serves as the bit line **131**, and the patterned conductive material forms the bit line contact plug **133**. In addition, it is also possible to adaptively deposit bit line spacers **134**. With reference to FIGS. 6A-6C, FIG. 6A shows a schematic top structural view of bit line structures, FIG. 6B shows a schematic cross-sectional structural view along a line A-A' in FIG. 6A, and FIG. 6C shows a schematic cross-sectional structural view along a line B-B' in FIG. 6B. The bit line structure **13** may have the same line width as the bit line contact

plug **133** in the direction **D2**. Each adaptively deposited bit line spacers **134** may also fill in an air gap between the bit line contact plugs **133** and the bit line contact holes **133'**.

(64) Step **S402** may specifically involve: etching the substrate **10** between adjacent bit line structures **13** by means of a wet etching process or a dry etching process to form a contact opening that extends into the substrate **10** and at least expose part of the active region **11** and part of the shallow trench isolation region **12**. Referring particularly to FIGS. 7A-7C, FIG. 7A shows a schematic top structural view of a semiconductor device provided with a contact opening, FIG. 7B shows a schematic cross-sectional structural view along a line A-A' in FIG. 7A, and FIG. 7C shows a schematic cross-sectional structural view along a line B-B' in FIG. 7A. A line width of the contact opening in the direction **D1** may be a distance between the bit line spacers **134** of adjacent bit line structures **13**. In another embodiment, the line width of the contact opening in the direction **D1** may also be smaller than the distance between the bit line spacers **134** of adjacent bit line structures **13**.

(65) Step **S403** may involve the following processes. Specifically, isolation material layers **14'** are deposited in the contact opening, where the isolation material layers **14'** may be at the same height as the bit line spacers **134** on top surfaces of the respective bit line structures **13**. A mask layer covering the bit line spacers **134** and the isolation material layers **14'** is deposited and patterned. The patterned mask layer is then used as a mask to etch the isolation material layers **14'**, so as to form the isolation fences **14** that are arranged at intervals in the first direction. Adjacent bit line structures **13** in the direction **D2** and adjacent isolation fences **14** in the direction **D1** may isolate the contact opening into a plurality of contact holes **15**. Referring particularly to FIGS. 8A-8C, FIG. 8A shows a semiconductor device provided with contact holes **15**, FIG. 8B shows a schematic cross-sectional structural view along a line A-A' in FIG. 8A, and FIG. 8C shows a schematic cross-sectional structural view along a line B-B' in FIG. 8A.

(66) The isolation material layers **14'** may be insulation material layers. As a specific example, the isolation material layers **14'** may be silicon nitride layers.

(67) Step **S404** may specifically involve: performing pretreatment on a bottom part of each of the contact holes **15** by using a germane gas. The active region **11** exposed in the contact holes **15** is extremely easily oxidized to form an insulation layer. For example, when the substrate **10** is a silicon layer, silicon oxide would be formed on an upper surface of the active region **11**, and germanium ions in germane can damage a silicon-oxygen bond, to form volatile germanium oxide, such that the insulation material of the active region **11** can be cleaned, thereby facilitating great conductivity between each of the contact pads and the active region **11**.

(68) Step **S405** may specifically involve: forming, by means of a chemical vapor deposition process or a physical vapor deposition process, the first contact pad **161** in a respective contact hole **15** that has been subjected to the pretreatment. It is possible to control, by means of selecting a suitable deposition condition, the first contact pad **161** to be in contact with part of a substrate surface of the respective contact hole **15**. As one example, the first contact pad **161** may be in contact with part of a surface of each of the active region **11** and the shallow trench isolation region **12** in the respective contact hole **15** (not shown in the figures). As another example, the first contact pad **161** may be formed, by a selective epitaxial growth process, on the active region **11** in the respective contact hole **15** that has been subjected to the pretreatment. The first contact pad **161** is only in contact with the active region **11**. As a specific example, the first contact pad **161** may be formed, by using the selective epitaxial growth process with silicane and germane as a precursor, on the active region **11** in the respective contact hole **15** that has been subjected to pretreatment. By using the selective epitaxial growth process, a bottom part of the first contact pad **161** is only in contact with the active region **11**. Referring particularly to FIGS. 9A and 9B, FIG. 9A shows a schematic cross-sectional structural view, along the line A-A', of a semiconductor device provided with first contact pads **161**, and FIG. 9B shows a schematic cross-sectional structural view, along the line B-B', of a semiconductor device provided with first contact pads **161**.

(69) Step **S406** may specifically involve: forming, on the first contact pad **161** and by using the

selective epitaxial growth process, the second contact pad **162** adaptively covering the first contact pad **161**. The second contact pad **162** and the first contact pad **161** have different chemical compositions. For example, the second contact pad **162** may be formed from a material different from that of the first contact pad **161**, and the second contact pad **162** may also be formed from the same material as that of the first contact pad **161** but with a different ion doping concentration. As a specific example, the second contact pad **162** adaptively covering the first contact pad **161** is formed on the respective first contact pad **161** by using the selective epitaxial growth process with silicane and germane as a precursor. Referring particularly to FIG. **10A**, there is shown a schematic cross-sectional structural view, along the line A-A', of a semiconductor device provided with second contact pads **162**. As an example, an upper surface of each of the second contact pads **162** may be higher than a lower surface of each of the bit lines **131**. FIG. **10B** also shows a schematic cross-sectional structural view, along the line B-B', of a semiconductor device provided with second contact pads **162**. After each second contact pad **162** is formed by using the selective epitaxial growth process, an air gap may be formed between each of the first contact pad **161** and the second contact pad **162** and a side wall on a side of the respective contact hole **15**.

(70) It should be noted that since the contact hole **15** has a larger aspect ratio and the first contact pad **161** is in contact with part of the substrate surface of the respective contact hole **15**, an air gap will be formed between the first contact pad **161** and each of a side wall on a side of the respective contact hole **15** and a bottom surface of the respective contact hole **15** after the first contact pad **161** and the second contact pad **162** are both formed, but such an air gap may not be able to be completely filled when the contact plugs **163** are subsequently formed by using a depositing process.

(71) As an example, step **S407** may specifically involve: forming a respective contact plug **163** on a respective second contact pad **162** by means of a chemical vapor deposition process or a physical vapor deposition process. Process conditions are controlled in such a way that the contact plug **163** only covers part of the surface of the respective second contact pad **162**, and the air gap **17** is formed between each of the first contact pads **161** and the second contact pads **162** and the respective contact hole **15**, as depicted in FIG. **11B** for details. As another example, step **S407** may specifically involve: forming a respective contact plug **163** on a respective second contact pad **162** by choosing a depositing process with a better step coverage characteristic, such that the contact plug **163** covers the second contact pad **162**, and is in contact with part of the respective first contact pad **161**, the air gap between each of the first contact pad **161** and the second contact pad **162** and the respective contact hole **15** is partially filled, and the air gap **17** is still reserved between the first contact pad **161** and a bottom part of the respective contact hole **15**, as depicted in FIG. **2B** in Embodiment One for details. Hereinafter, descriptions will be made based on the example that a contact plug **163** covering part of a surface of a respective second contact pad **162** is formed in a respective contact hole **15**. With reference to FIGS. **11A-11C**, FIG. **11A** shows a schematic top structural view of a semiconductor device provided with contact plugs **163**, and FIG. **11B** shows a schematic cross-sectional structural view along a line A-A' in FIG. **11A**. An upper surface of a first contact insertion plug **1631** may be higher than an upper surface of a bit line **131**, so that a contact interface between the first contact insertion plug **1631** and a second contact insertion plug **1632** is higher than the bit line **131**, thereby avoiding ions at the contact interface from penetrating into the bit line **131** and affecting device performance. FIG. **11C** shows a schematic cross-sectional structural view along a line B-B' in FIG. **11A**.

(72) Step **S407** may also specifically involve: depositing, by means of the chemical vapor deposition process or the physical vapor deposition process, a polysilicon layer, a metal silicide or a metal layer in a respective contact hole **15** as its first contact insertion plug **1631**; and depositing a metal layer in the respective contact hole **15** to form its second contact insertion plug **1632** covering the first contact insertion plug **1631**. The first contact pad **161**, the second contact pad **162**, the first contact insertion plug **1631** and the second contact insertion plug **1632** may form the

contact pad structure **16**.

(73) As an example, the first contact insertion plug **1631** may be a silicon phosphide layer, and the second contact insertion plug **1632** may be made of tungsten. When the first contact insertion plug **1631** is made of the same material as the second contact pad **162**, the growth rate of the first contact insertion plug **1631** may be greater than that of the second contact pad **162**.

(74) It should be noted that after the polysilicon layer, the metal silicide or the metal layer is formed in the respective contact hole **15** as the first contact insertion plug **1631**, a heavily-doped semiconductor layer **18** covering the first contact insertion plug **1631** may also be deposited.

(75) The above embodiment provides a method for preparing a semiconductor device. In this method, pretreatment is performed on a bottom part of each contact hole **15** by using a germane gas, first contact pads **161** and second contact pads **162** adaptively covering the respective first contact pads **161** grow on an active region **11** by using a selective epitaxial growth process, to form contact plugs **163** in the contact holes **15**, respectively, such that the first contact pads **161** are in full contact with the active region **11** in a substrate **10**, which can effectively reduce contact resistance. Moreover, by providing contact pads in the form of a stacking structure, each of the first contact pads can function as a buffer layer for a respective second contact pad **162** so as to avoid doped ions in the second contact pad **162** from penetrating into the substrate **10** and affecting conductivity. In addition, an air gap **17** in each contact hole **15** is formed between the first contact pad **161** and a side wall on a side of the contact hole **15**, which can effectively reduce an influence of the air gap on the conductivity, thereby facilitating significant improvement of a conductivity characteristic of the semiconductor device.

(76) Although various implementations are disclosed herein, it should be understood that they have been presented for facilitating the understanding of the present disclosure, rather than limiting the scope of the present disclosure. A skilled person in the art can make any modification and change to the form and details of the implementation without departing from the spirit and scope of the present disclosure, and the scope of the present disclosure shall be defined in the appended claims.

Claims

1. A method for preparing a semiconductor device, comprising: forming bit line structures extending in a first direction on a substrate, wherein the substrate comprises at least an active region and a shallow trench isolation region, the bit line structures are arranged at intervals in a second direction, and the second direction is perpendicular to the first direction; etching the substrate between adjacent bit line structures to form a contact opening that extends into the substrate and at least expose part of the active region and part of the shallow trench isolation region; forming isolation fences in the contact opening, wherein the isolation fences are arranged at intervals in the first direction to isolate the contact opening into a plurality of contact holes; performing pretreatment on a bottom part of each of the contact holes; forming a plurality of first contact pads in the respective contact holes; forming a second contact pad adaptively covering the first contact pad, wherein an air gap is formed between the first contact pad and a side wall on a side of the respective contact hole; and forming contact plugs in the contact holes, respectively.
2. The method according to claim 1, wherein the first contact pad is in contact with part of a substrate surface.
3. The method according to claim 2, further comprises: forming the first contact pad by using a selective epitaxial growth process.
4. The method according to claim 1, further comprises: forming the second contact pad adaptively covering the first contact pad, wherein the air gap is among of first contact pad, the second contact pad and the side wall on a side of the respective contact hole.
5. The method according to claim 1, further comprises: forming a contact plug covering the respective second contact pad and in contact with part of a surface of the respective first contact

pad.

6. The method according to claim 1, further comprises: forming a polysilicon layer, a metal silicide or a metal layer in the respective contact hole as first contact insertion plug, and depositing a metal layer to form second contact insertion plug covering the first contact insertion plug.

7. The method according to claim 6, wherein the first contact insertion plug is made of the same material as the second contact pad, and a growth rate of the first contact insertion plug is greater than a growth rate of the second contact pad.

8. The method according to claim 6, wherein after the step of forming the polysilicon layer, the metal silicide or the metal layer in the respective contact hole as the first contact insertion plug, the method further comprises: depositing a heavily-doped semiconductor layer covering the first contact insertion plug.

9. The method according to claim 6, wherein an upper surface of the second contact pad is higher than a lower surface of the bit line, and an upper surface of the first contact insertion plug is higher than an upper surface of the bit line.

10. The method according to claim 1, further comprises: depositing a conductive layer covering the substrate and a hard mask layer covering the conductive layer; and patterning the conductive layer and the hard mask layer, wherein the patterned conductive layer serves as a bit line, and the patterned hard mask layer is located on the bit line.
